0055. Jump Game

Medium · Greedy

leetcode.com/problems/jump-game

Problem

- Nums[i] is the maximum jump length from i. Starting at index 0, can you reach the last index?

Intuition

- Track the farthest index reachable so far. If the current index is beyond it, we are stuck.

Brute Force

- DP/DFS over every jump: O(n^2).

Optimal Approach

- Reach = 0.
- For i: if i > reach return false; reach = max(reach, i + nums[i]).
- Return true.

Worked Example

```
[3,2,1,0,4].

i=0 reach=3
i=1 reach=3
i=2 reach=3
i=3 reach=3
i=4 > reach 3 -> false
```

Java Solution

```java
public class JumpGame {
    public boolean canJump(int[] nums) {
        int reach = 0;
        for (int i = 0; i < nums.length; i++) {
            if (i > reach) return false;
            reach = Math.max(reach, i + nums[i]);
        }
        return true;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- Early exit once reach >= n-1 is an optional speed-up.